

Predicting Next-Night Sleep Quality from DaytimeWearable Physiology and Activity: A Single-Subject(N-of-1) Machine-Learning Study

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Introduction

Sleep quality is shaped by both continuity and architecture, and is closely tied to the autonomic nervous system. Consumer wearables offer non-invasive access to heart-rate variability, heart rate, SpO₂ and activity, raising the question of whether daytime physiology can forecast the following night's sleep in a single individual.

Method

We conducted a retrospective, single-subject (N-of-1) study using five months of consumer wearable exports (25 Nov 2025 – 4 May 2026). Sleep sessions, daily vital signs (HR, SpO₂, HRV) and daily activity (steps, calories) were cleaned, aggregated to daily records, and filtered for outliers using the 1.5×IQR rule. An inner join across streams produced **124 complete person-days** with no missing values.

To capture intra-individual shifts, each physiological and activity measure was expressed as its deviation from a personal 7-day rolling mean. The final predictor set contained six variables: relative HRV, HR, steps, calories, SpO₂, and same-night sleep latency. Three outcomes were modelled: Sleep Time Ratio (efficiency), Deep Sleep duration, and WASO (fragmentation).

Data were split chronologically (80/20; 99 train, 25 test days) and standardised using training-set parameters only. A **Random Forest** was tuned via grid search with 5-fold TimeSeriesSplit cross-validation (negative MAE objective). For comparison, an **LSTM** (50 units, dropout 0.2, 3-day window, Adam, MSE, 50 epochs) was trained on min-max scaled inputs. Uncertainty was quantified via 1,000-sample bootstrap 95% CIs. Interpretability used Gini importance and SHAP, following TRIPOD+AI and CLIX-M reporting checklists.

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Data Science and AI

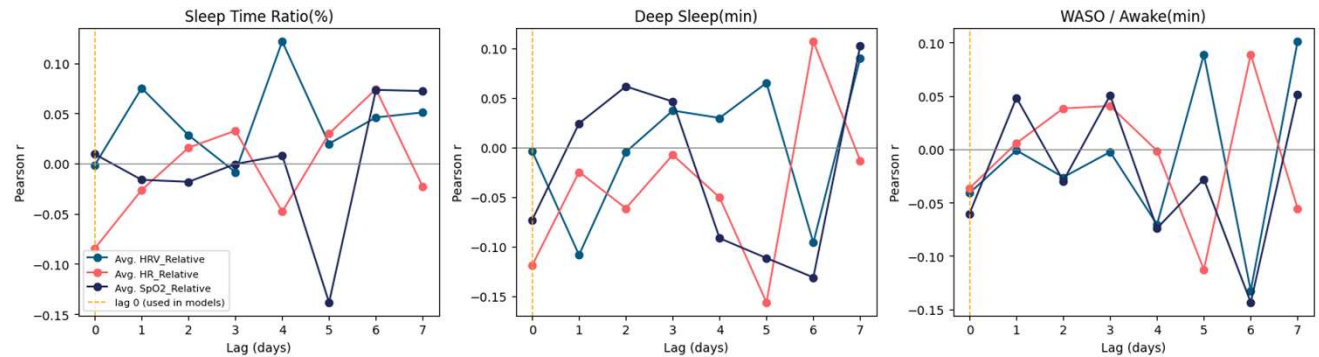


Figure 2. Lagged Pearson correlations of relative HRV, HR and SpO₂ with each sleep outcome at lags 0–7 days. All associations weak ($|r| < 0.14$) and non-significant.

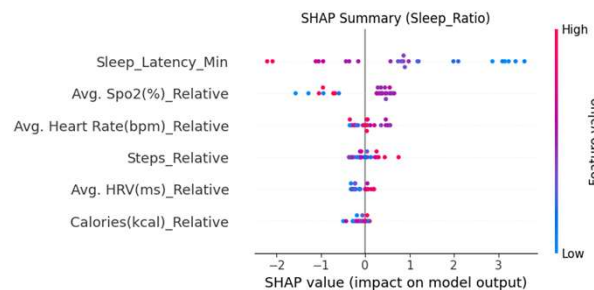


Figure 1. SHAP summary for the Sleep Time Ratio Random Forest. Sleep latency dominates — higher latency lowers predicted efficiency — while physiological features contribute weakly and inconsistently.

Results

Final sample: 124 person-days. All pairwise predictor correlations $|r| < 0.40$.

RQ1 (predictability). For sleep efficiency, Random Forest achieved MAE = 3.06 pp, $R^2 = 0.29$ (bootstrap 95% CI [2.32, 3.96]). Deep Sleep and WASO were not predictable (RF $R^2 = -0.07$ and -0.26). Random Forest outperformed LSTM on every outcome; the LSTM failed to generalise.

RQ2 (importance, Figure 1). Same-night sleep latency dominated (≈ 0.32), roughly double any other feature. SHAP: higher latency → lower predicted efficiency; physiological features small and inconsistent.

RQ3 (lags, Figure 2). All HRV–sleep correlations at lags 0–7 days were weak and non-significant (strongest $r = 0.122$ at lag 4). No evidence HRV precedes poor sleep.

Conclusions

In this N-of-1 study, sleep efficiency was modestly but reliably predictable ($R^2 = 0.29$), while deep sleep and fragmentation were not. The efficiency signal was driven by sleep latency, not autonomic physiology, and HRV showed no association with sleep at any lag up to one week. Random Forest outperformed LSTM in this low-N regime. Daytime HRV does not function as an early marker of poor sleep in this individual.

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